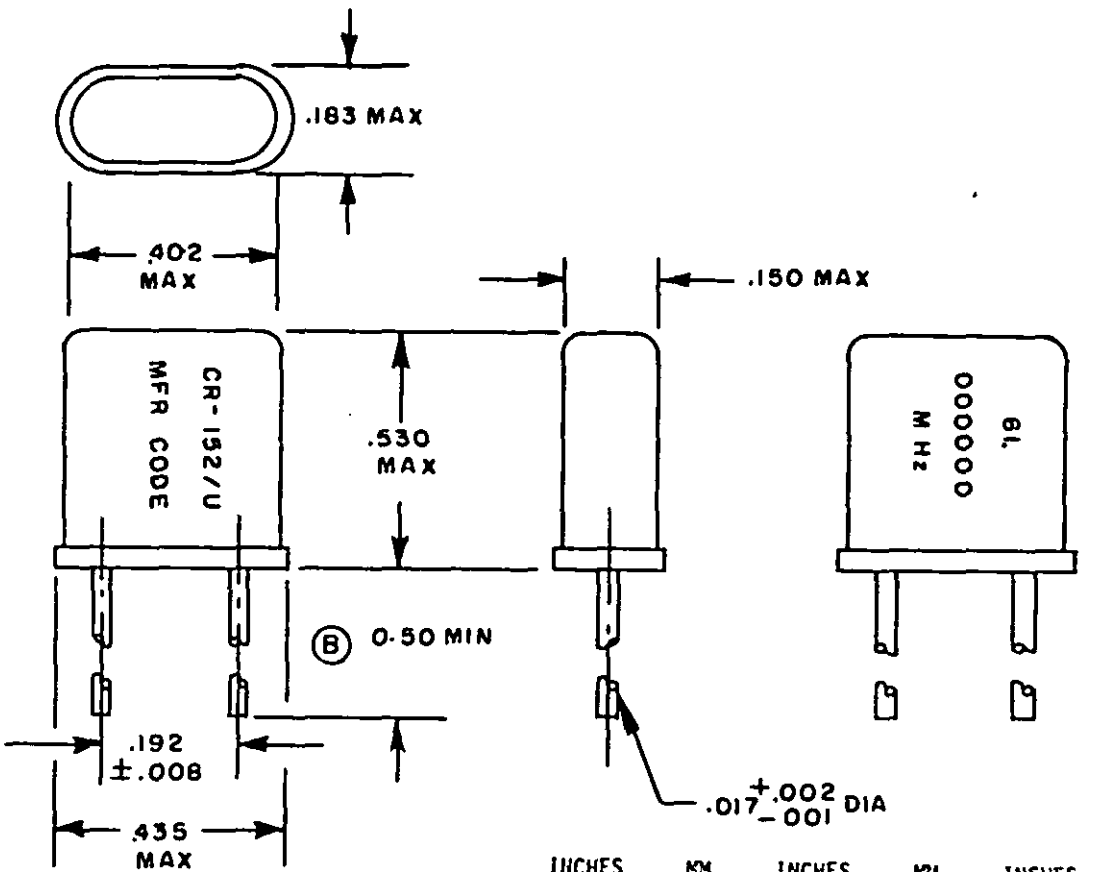
MILITARY SPECIFICATION SHEET

CRYSTAL UNIT, QUARTZ, CR-152/U

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for acquiring the crystal unit described herein shall consist of this specification and the latest issue of MIL-C-3098.

Pertinent characteristics: 16 to 61 MHz; third mechanical overtone; noncontrolled; series resonance.



INCHES	MM	INCHES	MM	INCHES	MM
.001	.03	.150	3.81	.435	11.05
.002	.05	.183	4.65	.50	12.7
.008	.20	.192	4.88	.530	13.46
.017	.43	.402	10.21		

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only and are based upon 1.00 inch = 25.4 mm.
3. The type-designation, frequency, and code-designating characters shall be not less than .038 (.97 mm) in height and not less than .027 (.69 mm) in width.

FIGURE 1. Dimensions and configuration.

Ⓑ denotes changes

FSC 5955

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TABLE I. Requirements.

Requirements	Specified value or condition
(B) Holder	HC-18/U or HC-43/U (MIL-H-10056/9)
Dimensions, marking, and configuration	See figure
Frequency range	16 to 61 MHz, inclusive
Capacitance, shunt	3.5 pF, maximum
Mode of oscillation	Third mechanical overtone
Government reference standard test set	TS-683/TSM
Rated drive level	2.0 mW, maximum
Calibration values:	
Resistance	40 ohms
Resistor voltage drop	0.28 volt
Operating temperature ranges:	
Primary	-40°C to +90°C, inclusive
Secondary	-55°C to -40°C and 90°C to 105°C, inclusive
Frequency tolerances:	
Primary operating temperature range	±20 PPM
Secondary operating temperature range	±30 PPM
Equivalent resistance	40 ohms, maximum
Unwanted modes	Method II, minimum unwanted-to-main mode resistance ratio of 3:1
Shock:	
Frequency change permitted	±5 PPM
Equivalent resistance change permitted	±10 percent
Vibration: 1/	
Frequency change permitted	±5 PPM
Equivalent resistance change permitted	±10 percent
Thermal shock:	
Frequency change permitted	±5 PPM
Equivalent resistance change permitted	±10 percent
Aging:	
Frequency change permitted	±5 PPM

1/ Method 204 of MIL-STD-202, test condition A.

NOTE: These crystal units are for use in the AN/ARC-101 equipment. They were previously fabricated using AVCO Corp. Specification 809051-1 through -10.

Custodians:
Army - ER
Navy - EC
Air Force - 85

Review activities:
Army - MI, AR, SM
Navy - SH
Air Force - 11, 99
DLA - ES

Users activities:
Navy - AS, CG, MC, OS
Air Force - 17, 19

Agent:
DLA - ES

Preparing activity:
Army - ER

(Project 5955-0623-12)

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